IMPLEMENTATION PLAYBOOK / 01

Group workflow and delivery plan

Interrupt-Driven Sensor Arbitration on ZedBoard

Hardware: ZedBoard + PMOD ACL2 + PMOD HYGRO + PMOD OLED
Planning window: 27 October - 20 November 2026
Target: integrated demonstration by 15 November; submission on 20 November.

What the team will build

A custom RTL controller accepts accelerometer interrupt requests and timer-generated temperature/humidity requests, schedules bounded bus transactions with configurable priority and aging, buffers completed records, and shares OLED and UART output fairly. A synthetic traffic mode makes contention repeatable for latency and starvation experiments.

Hardware facts that determine the design

Device	Verified capability	Design consequence
ACL2	ADXL362; SPI; INT1/INT2 can signal data-ready or FIFO conditions. [1,4]	Use a genuine external interrupt for acceleration acquisition.
HYGRO	HDC1080; I2C at 7-bit address 0x40; no interrupt output. [2,5]	Use a sampling timer and conversion-wait event; do not claim a sensor-generated interrupt.
OLED	128 x 32 SPI display; requires power/reset sequencing. [3]	Use a dedicated display driver and bounded frame updates.
ZedBoard UART	Built-in USB-UART connects to PS UART1, MIO 48/49. [6]	Use a small bare-metal PS application as the UART bridge; a PL TX wire cannot directly reach that port.

Scope and assumptions

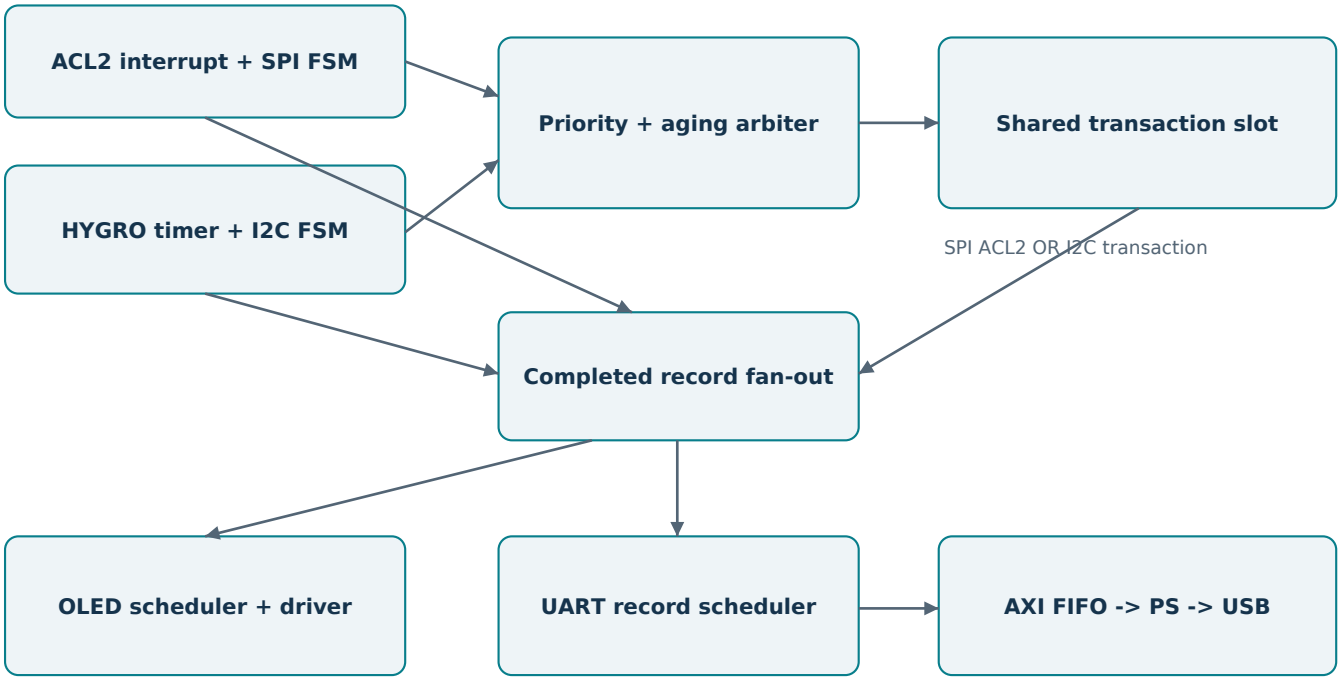
The plan assumes four work packages, not a known group size. Assign names on day one; combine packages for three members or split verification for five. Use SystemVerilog, Vivado, and a matching Vitis/SDK release supported by your lab. Existing IP may provide AXI infrastructure; the arbitration policy and sensor control remain your RTL.

Terminology: these sensors are bus slaves. This is a multi-requester service arbiter, not electrical I2C multi-master arbitration. If the original title is fixed, explain this distinction in the report. A descriptive title is **Interrupt- and Timer-Event-Driven Sensor Arbitration with Aging on ZedBoard**.

This document is a proposed implementation contract, not completed HDL or measured performance. Register offsets and module names below are project choices; chip behavior is grounded in the references on page 12.

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Architecture to implement



Acquisition and display use separate physical SPI engines.

Three scheduling boundaries

• **Acquisition:** two logical requesters compete for one transaction slot. The grant selects the corresponding SPI or I2C engine. This deliberate serialization defines the resource whose fairness is measured.

• **UART:** completed ACL2/HYGRO records compete for a UART record queue. Grant one complete record; the PS is the only physical UART writer, so text cannot interleave.

• **OLED:** each source updates its own latest-value mailbox. An arbiter selects a pending source to update a back buffer. A complete snapshot is frozen during each display transfer.

Critical architectural rules

Do not hold the acquisition slot during HYGRO conversion. Treat conversion start and result read as separate transactions, and age only requests that are eligible to run. Once a physical transaction begins, do not preempt it. Give every transaction a timeout so a faulty slave cannot lock the scheduler.

Use one PL system clock, preferably PS FCLK at 100 MHz, for control, AXI, and FIFOs. Generate SPI/I2C timing with clock-enable pulses rather than using divided fabric clocks. Synchronize external inputs, and synchronize reset deassertion. Confirm timing closure at the chosen frequency before freezing it.

Acquisition and output must not share a blocking handshake. If UART buffers fill, count and drop the newly completed UART record under the baseline policy; still update the OLED mailbox. Display coalescing is intentional and is counted separately from sample loss.

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Repository structure and build order

Create one shared Git repository. Commit source, constraints, scripts and small evidence files. Keep generated tool directories and large run outputs outside version control.

```
zedboard-arbiter/
  README.md                # setup, build and demonstration
  rtl/
    common/                # config package, reset, sync, FIFO
    control/               # pending requests, aging, dispatcher
    sensors/               # ACL2 and HYGRO service FSMs
    bus/                   # SPI and I2C byte/transaction engines
    output/                # record fan-out, UART/OLED scheduling
    metrics/               # counters and synthetic traffic
    axi/                   # custom AXI4-Lite register interface
    sensor_hub.sv          # portable custom RTL integration
    zedboard_top.sv         # board / block-design wrapper
  assets/font8x8.mem        # OLED character bitmap ROM
  constraints/zedboard_pmods.xdc
  sim/                     # unit TBs, device models, system TB
  sw/                      # bare-metal PS UART application
  scripts/                 # reproducible Vivado/sim/analysis
  docs/                    # contracts, wiring, test/demo plan
  results/                 # approved CSV logs, plots, summaries
  build/                   # ignored generated Vivado/Vitis files
```

Build in this order

Step	Required files first	Why this order matters
1	config_pkg.sv, interfaces.md, register_map.md, wiring.md	All members agree on widths, handshakes, ownership and pins before coding.
2	priority_aging_arbiter.sv + tb_arbiter.sv; SPI/I2C engines + unit TBs	Policy and bus engines can progress independently.
3	acl2_service.sv, hygro_service.sv + device models	Sensor FSMs depend on tested bus contracts, not ad hoc signals.
4	record_fanout.sv, output schedulers, AXI registers, PS bridge	Use synthetic records before real sensor integration.
5	sensor_hub.sv, block-design Tcl, board wrapper, XDC	Integrate verified blocks with stable interfaces.
6	stress TB, collection/analysis scripts, final report	Measure only the configuration that will be demonstrated.

Files generated by tools

Vivado generates the block-design HDL wrapper, implementation outputs, bitstream and hardware handoff (.xsa, or .hdf in older flows). Vitis/SDK generates the platform/BSP and application ELF. Reproduce them from scripts and recorded tool versions; do not hand-edit generated wrappers. A packaged custom AXI IP also needs tool-generated component.xml and packaging metadata.

Use one top-level configuration for clock frequency, SPI/I2C divisors, sampling interval, FIFO depths and timeout lengths. Simulation may override time constants, but the hardware build must restore real sensor delays.

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RTL file inventory: common and acquisition

Each row below is a required source file in the proposed split. Small files may be combined only after the team agrees and updates the interface document.

Path under rtl/	Purpose and reason needed	By
common/config_pkg.sv	Defines record/command types, source IDs, widths, parameters and shared constants. Prevents incompatible definitions between members.	A
common/reset_sync.sv	Asserts reset safely and synchronizes release in the PL clock domain; avoids partially released state machines.	A
common/sync_input.sv	Two-stage synchronizer for ACL2 interrupt and asynchronous controls. Mark synchronizer stages appropriately.	A
common/sync_fifo.sv	Reusable single-clock FIFO with valid/ready and occupancy; absorbs finite bursts. Define simultaneous push/pop and full behavior.	A
control/event_pending.sv	Stores one outstanding request per source and records its initial timestamp. New events must win over a same-cycle clear; repeated events are coalesced and counted.	A
control/priority_aging_arbiter.sv	Reusable two-input arbiter. Supports fixed priority and priority plus aging, fair ties and bounded grant ownership. Instantiate separately for acquisition and outputs.	A
control/transaction_dispatch.sv	Accepts one granted sensor transaction, starts the selected engine, routes its response and releases the slot on success/error/timeout.	A
bus/spi_master.sv	Reusable SPI transaction engine; preserves chip select across command/address/data. Instantiate separately for ACL2 and OLED.	B
bus/i2c_master.sv	Implements I2C transfers, ACK/NACK, STOP, line release and timeout. Open-drain SDA/SCL must drive 0 or high impedance, never logic 1.	C
sensors/acl2_service.sv	Initializes ADXL362, verifies identity, maps data-ready interrupt, performs coherent XYZ burst reads and forms records. Handles interrupt rearm.	B
sensors/hygro_service.sv	Initializes HDC1080; schedules sample, triggers conversion, waits without holding the slot, then reads four bytes and forms a record.	C
sensor_hub.sv	Connects custom RTL, shared transaction slot, outputs and metrics. Used in system simulation without the Zynq processing system.	A
zedboard_top.sv	Connects block-design wrapper, actual board pins and PMOD I/O buffers. Keeps board-specific wiring out of the portable hub.	A

Ownership key: A = architecture/arbitration; B = ACL2/SPI; C = HYGRO/I2C; D = output/PS. These are work packages, not required group membership counts.

IMPLEMENTATION PLAYBOOK / 05

RTL and software inventory: outputs and observability

Path	Purpose and reason needed	By
rtl/output/record_fanout.sv	Copies each completed record into its source UART FIFO and latest OLED mailbox independently. Separates overflow from display coalescing.	D
rtl/output/uart_record_mux.sv	Uses the shared arbiter module to move one complete source record into the PS-facing FIFO. Holds ownership until that record is enqueued.	D
rtl/output/oled_update_mux.sv	Arbitrates dirty source mailboxes and updates a back buffer. Keeps selected source data stable during formatting.	D
rtl/output/value_format.sv	Converts raw values into signed decimal/fixed-point glyph indices for display; avoids floating point in RTL.	D
rtl/output/oled_driver.sv	Sequences display power, reset and commands, then sends immutable 512-byte frame snapshots through a dedicated SPI instance.	D
rtl/output/font_rom.sv + assets/font8x8.mem	Maps character codes to pixel columns. With an 8 x 8 font the display fits 16 columns by 4 rows.	D
rtl/metrics/metrics.sv	Counts requests, completions, wait sums/maxima, errors, overflows and coalescing. Provides coherent snapshots for reporting.	A
rtl/metrics/traffic_gen.sv	Generates repeatable contention and bounded mock service times. Enabled explicitly in test mode; never drives synthetic traffic onto sensor pins.	A
rtl/axi/axi_regs.sv	AXI4-Lite slave exposing configuration, metrics and a record FIFO. Decouples PS output transport from PL sensor service.	D
sw/main.c	Initializes platform/UART/GIC, configures PL, receives record notifications and outputs tagged lines. Does not poll sensor registers.	D
sw/pl_regs.h	Matches the agreed RTL register offsets and field definitions. Prevents software/hardware address mismatches.	D
sw/pl_irq.c + sw/pl_irq.h	PL-to-PS interrupt setup, FIFO draining and software ring buffer. Records software overflow and keeps ISR bounded.	D

Keep the PS bridge small

The PL IRQ is level-high while the PS-facing record FIFO is nonempty or an enabled sticky error is set. The ISR copies complete records into a software ring and returns. Main code formats and sends complete lines using PS UART1. Waiting for UART TX space is acceptable in this output bridge; it must not block PL acquisition or keep the ISR occupied. Idle main code can sleep until an interrupt.

Mask/acknowledge sources according to their semantics. Popping the final FIFO record deasserts its level interrupt; a sticky error is cleared explicitly. If the software ring fills, count the loss and drain/discard rather than leaving the interrupt permanently asserted. With a budgeted ISR, use a deferred-drain path that resumes before unmasking.

No separate PL uart_tx.sv is required for this baseline: the onboard USB-UART uses the PS peripheral. An all-PL UART is an optional alternative only if an external USB-to-TTL adapter is actually available.

IMPLEMENTATION PLAYBOOK / 06

Interface contract: agree before parallel coding

Clock, reset and transaction ownership

Use clk and active-high synchronous rst internally. All ready/valid transfers occur on a rising edge when both are 1. The sender holds valid and payload stable while stalled. A command is accepted exactly once; each accepted command produces exactly one response. Do not use one-cycle done pulses without a guaranteed receiver.

Boundary	Proposed fields	Required behavior
Service FSM -> dispatcher	cmd_valid, cmd_ready; source, bus, read/write, register address, byte count, write data	At most one outstanding command per source. The service FSM holds its command until granted/accepted.
Dispatcher -> service FSM	rsp_valid, rsp_ready; source, read data, error code	Transaction slot remains owned until engine stops safely; buffered response delivery may follow. No wrong-source response.
Service FSM -> fan-out	record_valid, record_ready; 256-bit record	One record per successful complete sensor acquisition. An ingress skid register absorbs the commit; output-full policy prevents indefinite acquisition stalls.
FIFO -> consumer	valid, ready, data; occupancy	Pop only on valid AND ready. Stable data during stall. Never overwrite unread entries.
Event -> arbiter	pending[1:0], eligible[1:0], base priority, aging enable	Only pending AND eligible requests compete. No aging while intentionally awaiting HYGRO conversion.
Arbiter -> dispatcher	grant_valid, grant_source; accept	Select when slot is idle; lock accepted transaction through completion. Age reset occurs on accepted grant, not merely selection.

Fixed record format: eight 32-bit words

W0: source ID and status; **W1:** per-source sequence number; **W2:** event timestamp; **W3:** acquisition-completion timestamp; **W4:** X[15:0] and Y[31:16]; **W5:** Z[15:0], upper half reserved; **W6:** temperature raw[15:0], humidity raw[31:16]; **W7:** accumulated transaction-wait cycles for this acquisition. Unused payload fields are zero. Keep raw sensor data for reproducible conversion.

Aging policy that is easy to verify

Use an explicit starvation override: fixed-priority selection normally; when aging is enabled and any eligible requester has waited AGE_LIMIT cycles, choose the oldest overdue requester. Break equal-age ties round-robin. Saturate age counters; never wrap. A newly serviced requester resets its age, allowing an older pending peer to win next.

This avoids a capped additive score that cannot overcome a configured priority gap. Define AGE_LIMIT, permitted priorities and tie semantics in interfaces.md. Bounded service and bounded downstream acceptance are prerequisites for any bounded-wait claim.

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AXI register map and Zynq integration

The following offsets are a proposed project register map, not vendor register addresses. All registers are 32-bit and offsets are relative to the assigned AXI base address. Allocate the address in Vivado and export it into pl_regs.h.

Offset	Name / access	Contract
0x00	CONTROL / RW	Bit 0 RUN; bit 1 AGING_EN; bit 2 SYNTH_MODE. Mode changes only while RUN=0 and engines are idle.
0x04	PRIORITY / RW	ACL2 base priority [7:0], HYGRO [15:8]. Larger value wins; apply same baseline policy to output schedulers.
0x08	AGE_LIMIT / RW	Wait threshold in PL cycles. Zero is illegal; clamp to 1 or reject and set config error.
0x0C	HYGRO_PERIOD / RW	Sampling period in cycles. Enforce documented minimum in real-sensor mode.
0x10	STATUS / RO	Init complete, engine busy, pending, test mode, error flags; exact bits frozen in register_map.md.
0x14	IRQ_ENABLE / RW	Enables FIFO-nonempty and sticky-error interrupt sources.
0x18	IRQ_STATUS / RO	Level sources plus sticky errors. Reading does not acknowledge or pop anything.
0x1C	ERROR_ACK / WO	Write-one-to-clear sticky errors; set wins over simultaneous clear.
0x20	FIFO_LEVEL / RO	Complete records in PS-facing FIFO; never a byte count.
0x24	FIFO_POP / WO	Writing bit 0 pops exactly one head record. Empty pop sets an error.
0x28	METRICS_CMD / WO	Bit 0 snapshot; bit 1 clear counters, permitted only when measurement is stopped.
0x40-0x5C	RECORD_W0..W7 / RO	Peek at head record. Head stays stable across eight reads until FIFO_POP; only the PS is allowed to pop.
0x80 onward	METRICS / RO	Snapshot bank: per-source accepted/completed/error counts, 64-bit wait sums, maxima, overflow/coalescing counts. Freeze full layout before implementation.

Vivado block design

Zynq Processing System M_AXI_GP0 -> AXI interconnect -> custom axi_regs peripheral. Use PS FCLK_CLK0 for the custom logic and AXI clock, with Processor System Reset for reset distribution. Route the PL level interrupt to IRQ_F2P and configure the PS GIC consistently. Enable UART1 on its board MIO route. Use the ZedBoard preset and verify it against the board guide. [6]

AXI implementation must accept AW and W independently, honor WSTRB, hold BVALID/RVALID until accepted, and prevent repeated side effects under stalls. Start from a vendor AXI4-Lite peripheral template rather than an unverified combinational register bus.

Checkpoint before sensors: PS writes/reads a scratch/config register, injects a synthetic record, receives an IRQ, reads eight words, pops once and prints one intact line. Prove this by 7 November.

IMPLEMENTATION PLAYBOOK / 08

Sensor, wiring and display bring-up procedure

1. Freeze wiring before attaching modules

Proposed allocation: **ACL2 -> JA**, **HYGRO -> JB**, **OLED -> JC**, subject to physical clearance and verification against your board revision. Use PL-connected headers, 3.3 V and common ground. Record connector orientation, PMOD pin number, signal name, FPGA package pin and I/O standard in docs/wiring.md. Copy package pins from the official ZedBoard master XDC/schematic; do not guess them.

ACL2: CS, MOSI, MISO, SCLK and INT1 must all be wired; a six-wire SPI-only connection omits the interrupt. HYGRO uses SCL and SDA with correctly enabled pull-ups. OLED additionally needs D/C, RES, VBATC and VDDC. Confirm all PMOD pinouts in [1-3]. Use safe idle outputs during reset; deassert all chip selects.

2. ACL2: identify -> configure -> interrupt -> burst read

Start SPI conservatively, for example 1 MHz, using the mode and edge timing in the ADXL362 datasheet. Verify identity, configure a moderate output rate such as 100 Hz and a known range, map data-ready to INT1, then enter measurement mode. Burst-read the six XYZ bytes and sign-extend the 12-bit values. Confirm stationary orientation and axis changes before enabling arbitration. [1,4]

Latch a synchronized interrupt once and rearm only after its previous assertion is cleared. Reading data clears data-ready, but deassertion may be delayed by up to 80 microseconds. Do not interpret a still-high line immediately after a read as a new sample. Use a rearm state and timeout; validate against the device model and ILA. A pending bit coalesces samples and cannot prove zero sensor loss; use ADXL362 FIFO/overrun tracking if lossless capture becomes required. [4]

3. HYGRO: trigger -> release slot -> timed wait -> read

At startup wait at least 15 ms, verify identity, set combined temperature/humidity acquisition, and use 7-bit address 0x40. Trigger with the temperature pointer, issue STOP, and release the shared slot. For 14-bit combined acquisition the manual lists 12.85 ms total typical conversion time; begin with a conservative 20 ms timer and handle a not-ready NACK with a bounded timed retry. Do not treat a typical time as a guaranteed maximum. Follow the manual recommendation to wait at least one full second between readings. [2,5]

Read four bytes, ACK the first three and NACK the last, then STOP. Convert unsigned 16-bit words using T = rawT x 165 / 65536 - 40 degrees C and RH = rawRH x 100 / 65536. Use sufficiently wide intermediate arithmetic for fixed-point display conversion. [2]

4. OLED: static screen before live screen

Implement the exact power/reset/command sequence from [3], clear memory, then show static text. Next implement four rows, for example X/Y, Z, temperature, humidity. Use a 512-byte immutable front buffer and a separate back buffer; swap only when the SPI engine is idle. Refresh at a proposed 5 Hz, coalescing intermediate display updates. UART may log more frequently than OLED.

IMPLEMENTATION PLAYBOOK / 09

Group ownership and dated execution plan

Package	Primary ownership	First independent proof
A: system + policy	Contracts, arbiter, dispatcher, pending events, metrics, integration, reproducible build	Arbiter TB with repeated high-priority requests and a bounded-wait low-priority requester.
B: ACL2 + SPI	SPI engine, accelerometer init/service, interrupt model and board validation	Identity and XYZ over SPI; real INT1 captured in ILA.
C: HYGRO + I2C	I2C engine, HYGRO FSM, conversion timer, sensor model and timeout tests	Identity and plausible raw temperature/humidity; correct ACK/NACK waveform.
D: outputs + PS	Record fan-out, UART/OLED schedulers, OLED formatting, AXI regs and PS bridge	Synthetic record printed via USB-UART; static OLED text.

For three members: combine B with A after SPI bring-up and split D between the other two. For five: give AXI/PS and verification/analysis a separate owner. Everyone writes tests for their own block; A coordinates system verification. The only physical board should have booked bring-up slots; simulation proceeds independently.

Dates	Parallel work and integration gate
27-28 Oct	All: freeze interfaces, register map, tool version and wiring. A creates common modules; B/C start bus engines; D proves PS UART and block design.
29-31 Oct	A: policy unit tests. B: ACL2 identity/XYZ. C: I2C model + identity. D: AXI register access and static OLED initialization.
1-3 Nov	B: real ACL2 interrupt. C: full timed HYGRO cycle. A: dispatcher/pending tests. D: synthetic record queues and UART line output. Gate: both sensors independently working.
4-7 Nov	Integrate acquisition, record fan-out and AXI IRQ path. Gate by 7 Nov: simultaneous live sensors produce tagged UART records; errors cannot deadlock the system.
8-11 Nov	Finish OLED live values, double buffering, output arbiters and metrics snapshots. Exercise configuration and capture ILA waveforms.
12-15 Nov	Run system stress, reset/backpressure/fault tests and hardware test mode. Fix defects; freeze features on 15 Nov.
16-17 Nov	Run matched fixed-priority/aging experiments, collect CSV, generate plots, interpret limitations.
18-19 Nov	Report, slides, demonstration video and clean rebuild on a second machine or clean directory. Preserve known-good bitstream and ELF.
20 Nov	Final demonstration and submission of the already tested build.

Daily integration rule: 15-minute check-in; each owner reports a demonstrated result, blocker and next handoff. Merge small tested changes daily. Do not wait for every block to be finished before connecting synthetic sources and sinks.

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Verification files and pass criteria

File under sim/	What it proves / why needed
tb_arbiter.sv	No grant without eligible request; at most one grant; configured priority; fair ties; aging override; no preemption; same-cycle arrival/accept behavior.
tb_fifo.sv	Ordering, depth boundaries, stable stalled data, simultaneous push/pop, full and empty cases.
tb_spi_master.sv	Correct clock phase/polarity, bit order, byte counts, continuous CS and response timing.
tb_i2c_master.sv	START/STOP, address direction, ACK handling, last-byte NACK, NACK/timeout release; line contention is never introduced by driving 1.
models/adxl362_model.sv	Identity/config registers, burst reads, sample changes and delayed interrupt clear. A model is needed before board access.
models/hdc1080_model.sv	Combined conversion delay, four-byte data, not-ready NACK and injected missing-device faults.
models/oled_monitor.sv	Checks command/data distinction, required startup sequencing, byte count and frame consistency; no need for a full graphical device model.
tb_sensor_services.sv	Correct initialization, ACL2 IRQ rearm and HYGRO slot release during conversion; accelerated simulation delays preserve sequencing.
tb_axi_regs.sv	Independent AW/W arrival, stalls, byte strobes, stable record peek, single explicit pop and correct level IRQ behavior.
tb_outputs.sv	Whole-record UART arbitration, output overflow policy, OLED coalescing and immutable front buffer.
tb_system_stress.sv	Portable sensor_hub with device models, randomized events, synthetic contention, backpressure, resets and a scoreboard.

System scoreboard and assertions

Track each accepted transaction by source and sequence. A response must match exactly one accepted command. Successful records remain in source order; losses must match explicit drop/coalescing counters. Grant vectors are one-hot-or-zero. An engine timeout must lead to released ownership. The scoreboard independently predicts policy decisions rather than copying the RTL implementation.

Required corner cases

Simultaneous requests; request during service; event at grant/completion; IRQ held high; ACL2 delayed deassertion; HYGRO not ready; absent I2C slave; saturated age counter; both FIFO boundaries; AXI stalls; OLED busy; PS not draining; reset during an active transfer. State what happens to partial records on reset: baseline discards all and begins a new run ID.

Hardware acceptance: both sensor values respond plausibly; no mixed UART lines or torn OLED frames; repeated resets recover; no unexpected errors in a proposed 10-minute nominal run; timing constraints are met. Simulated fairness passes do not establish hardware fairness until on-board counters or traces confirm the same behavior.

IMPLEMENTATION PLAYBOOK / 11

Measurement plan and efficient engineering rules

Measure distinct delays

Metric	Definition and interpretation
Transaction arbitration wait	Accepted command time minus first eligible-pending time. Excludes HYGRO conversion wait; sum trigger/read waits separately for a HYGRO acquisition.
Acquisition response	Record-completion time minus initial event time. Includes sensor conversion and bus transfer; not just arbiter performance.
Output queue wait	Output acceptance time minus record-ready time; count separately for UART and OLED. OLED latest-value coalescing changes its interpretation.
Service and loss	Accepted requests, successful completions, errors, FIFO high-water mark, UART drops, software-ring drops and display coalescing, per source.
Fairness under contention	Maximum eligible wait and service share under equal offered traffic; also show unequal-rate traffic. Equal counts are meaningful only for comparable demand and service cost.

Repeatable experiment matrix

Use identical seeds, request traces, service-time bounds, run duration, FIFO sizes and output conditions for policy comparisons. Baseline trials: (1) real sensors at nominal rates; (2) simultaneous synthetic requests; (3) continuously eligible high-priority stream plus a lower-priority stream; (4) output backpressure; (5) bus fault injection in simulation. Compare fixed priority with aging disabled and the same priorities with aging enabled.

Synthetic mode must bypass physical sensor transactions and label every log as SYNTHETIC. Real HYGRO cannot be accelerated arbitrarily to create stress. Export CSV columns: run_id, mode, seed, policy, clock_hz, age_limit, source, offered, accepted, completed, mean_wait_cycles, max_wait_cycles, errors, drops. Preserve the raw log alongside plots.

Bounded waiting: state the assumptions

With two requesters, an overdue low-priority request wins at the next available decision unless an older overdue peer wins first. Under continuous eligibility and a maximum accepted service duration Smax, a conservative bound for the proposed override is approximately AGE_LIMIT + 2 x Smax plus pipeline/response overhead. Derive the exact RTL-cycle bound and verify it; it is not a measured result. No finite bound applies when the required sink can remain stalled forever or offered load exceeds sustainable capacity without a defined drop policy.

Use 64-bit accumulation for wait sums and counter snapshots. A 32-bit 100 MHz cycle timestamp wraps in about 42.95 seconds: subtract modulo 2^32 only for intervals shorter than one wrap. Use a run epoch or wider time for longer intervals. Snapshot multiword counters before PS reads to avoid torn values.

Efficiency and risk priorities

Do first: contracts, synthetic traffic, independent sensor tests, PS UART bridge, live integrated acquisition. **Do next:** OLED and measured contention. **Defer:** Linux, DMA, a graphical UI, floating-point RTL, command parsers, shared physical SPI wiring, and sophisticated animations. If behind schedule, simplify display formatting and reuse validated SPI/IP infrastructure; keep the arbiter, tests and measurements as the central contribution.

At 115200 baud, 8N1 carries about 11,520 bytes/s. A 100 Hz ACL2 stream of 64-byte lines is about 6,400 bytes/s before extra traffic. Set a line-length budget, verify actual utilization, and decimate UART reporting explicitly if needed. Never confuse output decimation with sensor acquisition loss.

IMPLEMENTATION PLAYBOOK / 12

Supporting files, submission and first-day checklist

Required supporting file	Why it is needed
README.md; .gitignore	One entry point for setup/build/demo; excludes generated build caches and local paths.
docs/interfaces.md; register_map.md	Authoritative signal timing, record layout, policy semantics and PS/PL register contract.
docs/wiring.md; test_plan.md; demo.md	Verified connector/package-pin mapping, pass criteria, and a repeatable demonstration sequence.
constraints/zedboard_pmods.xdc	Package pins, I/O standards and justified timing constraints. Do not apply blanket false paths to hide violations.
scripts/create_project.tcl; create_bd.tcl; build.tcl	Recreates Vivado project, Zynq/AXI connections, custom IP packaging and bitstream build in a clean directory.
scripts/run_sim.tcl; collect_uart.py; analyze_results.py	Runs simulation regression, captures serial logs, calculates metrics and generates reproducible comparison plots.
results/manifest.csv; final_summary.md	Records exact commit, tool versions, bitstream/ELF identity, parameters, seeds and interpretation for every approved run.
Final report, slides, bitstream, ELF, XSA/HDF	Explains architecture and results; preserves a runnable demonstration with the matching hardware/software pair.

First working session: leave with these decisions made

1. Assign A/B/C/D owners and a backup integrator; book board access.
2. Verify board/PMOD revisions and connector orientation; create the wiring table.
3. Freeze clock, record layout, command/response handshake and register map.
4. Create repository and reproducible ZedBoard block design; prove USB-UART output.
5. Commit arbiter and device-model test scaffolds; use synthetic payloads for output work.
6. Set the next gate: independent ACL2/HYGRO bring-up, followed by live UART integration by 7 November.

Demonstration sequence

Boot the saved bitstream and ELF; print configuration and firmware/RTL version. Show live motion and environmental readings on UART/OLED. Change priority and aging configuration while stopped. Run labeled synthetic contention with fixed priority, then the identical trace with aging. Show measured lower-priority maximum wait and service progress. Restore real-sensor mode, demonstrate reset recovery, and retain the terminal log.

Manufacturer references

[1] Digilent PMOD ACL2 reference manual

[2] Digilent PMOD HYGRO reference manual

[3] Digilent PMOD OLED reference manual

[4] Analog Devices ADXL362 datasheet

[5] Texas Instruments HDC1080 datasheet

[6] ZedBoard Hardware User Guide v2.2

References identify manufacturer behavior; rates, FIFO sizes, register offsets, module boundaries and schedule in this plan are proposed project choices. Check timing and board connections against the actual module revisions before implementation.